

TEMPO-BIAS: Political Bias Analysis in LLMs

Research Methodology

Ran Lu, Hasti Hosseinpour, Abderrahmane Moujar, Oloruntobi Olutola, Jay Saumil Gupte

Paris-Saclay University
Fairness in AI Course under Prof. Adrian Popescu

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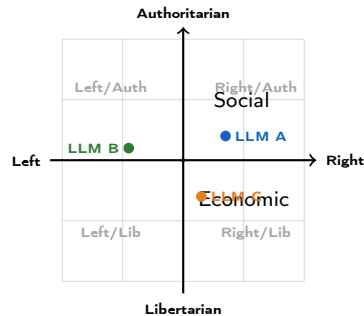
Research Context

Background:

- Large Language Models (LLMs) are increasingly used in society
- Training data and post-training alignment can embed biases
- Political bias can influence public opinion and democratic processes
- Need for systematic measurement and comparison across model families

Key Insight (Buyl et al., 2024):

"Large language models reflect the ideology of their creators"



Research Questions

RQ1: Bias Across Model Generations

How does political bias in LLM outputs change across successive model generations?

RQ2: Alignment and Red-Teaming Effects

To what extent do safety alignment and red-teaming interventions correspond to shifts in political stance and sentiment?

RQ3: Temporal and Ideological Asymmetry

Are temporal changes in political bias uniform across ideological dimensions and political entities, or do they exhibit asymmetric patterns?

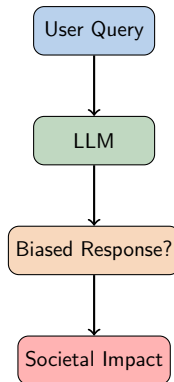
Why Study Political Bias in LLMs?

Societal Impact:

- LLMs used by millions daily
- Influence on information access
- Potential to shape political views
- Democratic implications

Research Gaps (from C2):

- Limited cross-model comparisons
- Pre vs. Post training understudied
- Lack of geographic diversity analysis
- Inconsistent measurement methodologies



Problem Formulation

Mathematical Framework:

Let $\mathcal{M} = \{M_1, M_2, \dots, M_n\}$ be a set of LLMs, and $\mathcal{Q} = \{q_1, q_2, \dots, q_m\}$ be a set of politically sensitive questions.

Definition 1 (Response Generation):

$$r_{i,j}^{(k)} = M_i(q_j, \theta, \tau) \quad \text{for run } k \in \{1, \dots, K\} \quad (1)$$

where θ represents model parameters and τ is the temperature.

Definition 2 (Sentiment Score):

$$s_{i,j}^{(k)} = \phi(r_{i,j}^{(k)}) \in [-1, 1] \quad (2)$$

where ϕ is a sentiment analysis function (VADER).

Definition 3 (Bias Score):

$$B_i = \frac{1}{|\mathcal{Q}_L|} \sum_{q \in \mathcal{Q}_L} \bar{s}_{i,q} - \frac{1}{|\mathcal{Q}_R|} \sum_{q \in \mathcal{Q}_R} \bar{s}_{i,q} \quad (3)$$

where $\mathcal{Q}_L$ and $\mathcal{Q}_R$ are left-aligned and right-aligned politician questions.

Bias Score Interpretation

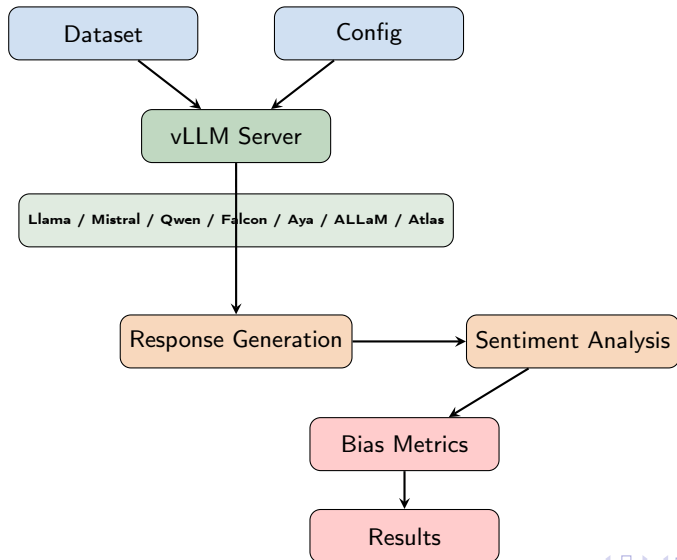
Interpreting the Bias Score B_i :

Score Range	Interpretation	Symbol
$B > 0.3$	Strong Left Bias	$\leftarrow\leftarrow$
$0.1 < B \leq 0.3$	Moderate Left Bias	$\leftarrow$
$-0.1 \leq B \leq 0.1$	Neutral / Balanced	$\circ$
$-0.3 \leq B < -0.1$	Moderate Right Bias	$\rightarrow$
$B < -0.3$	Strong Right Bias	$\rightarrow\rightarrow$

Statistical Significance:

- Multiple runs ($K \geq 3$) per question to assess variance
- Standard deviation σ_i measures response consistency
- T-tests for comparing pre vs. post-training models

Pipeline Overview



Data Selection & Collection

Multi-Source Dataset Strategy:

Dataset	Type	Size	Usage
Politicians Prompts	Custom	60+	Sentiment analysis on political figures
BABE (HuggingFace)	News Bias	4,000+	Media bias detection training
LLM Ideology Analysis	Ideology	2,500+	Cross-model ideology comparison
NewsMTSC (Hamborg)	News	11,000+	Multi-target sentiment classification
FreshBench (GJO)	QA	1,000+	Knowledge and opinion evaluation
VoteView	Congressional	50,000+	US political voting records
MAGPIE-BABE Model	Fine-tuned	–	Bias detection classifier

Primary Sources:

- **Political Compass:** 62 standardized questions (Economic + Social axes)
- **Politician Sentiment:** Left (Biden, Sanders) / Center (Merkel) / Right (Trump, Le Pen)

Dataset Details & References

Media Bias Datasets:

- **BABE** (Spinde et al.)
huggingface.co/datasets/mediabiasgroup/BABE
4,000+ annotated news sentences
- **NewsMTSC** (Hamborg et al., 2021)
aclanthology.org/2021.eacl-main.142/
Multi-target sentiment in news
- **MAGPIE-BABE Model**
huggingface.co/mediabiasgroup/magpie-babe-ft
Fine-tuned bias classifier

Key References:

- Cognitive biases in LLMs (PMC, 2023): pmc.ncbi.nlm.nih.gov/articles/PMC10071453/
- Political bias measurement (First Monday): firstmonday.org/ojs/index.php/fm/article/view/14491

Political & Ideology Datasets:

- **LLM Ideology Analysis**
huggingface.co/datasets/aida-ugent/llm-ideology-analysis
2,500+ ideology probes
- **VoteView**
voteview.com/data
US Congressional voting data
- **FreshBench**
github.com/FreedomIntelligence/FreshBench
LLM evaluation benchmark

7 Model Families Across Different Origins:

Family	Origin	Base Model	Post-trained
Llama 2/3	USA (Meta)	Llama-2-7B	Llama-2-7B-Chat
Mistral	France	Mistral-7B	Mistral-7B-Instruct
Qwen	China	Qwen-7B	Qwen-7B-Chat
Falcon	UAE	Falcon-7B	Falcon-7B-Instruct
Aya	Multilingual	–	Aya-101
ALLaM	Saudi Arabia	ALLaM-7B	ALLaM-7B-Instruct
Atlas	Morocco	North Africa	Atlas-Chat-9B

Selection Criteria:

- Geographic diversity (USA, Europe, Asia, Middle East, North Africa)
- Availability of both base and post-trained versions
- Comparable size (7-13B parameters)

VADER (Valence Aware Dictionary and sEntiment Reasoner):

Sentiment Extraction:

$$\phi(r) = \text{VADER}(r).\text{compound}$$
$$s \in [-1, 1]$$

- Rule-based sentiment analysis
- Designed for social media text
- Returns compound score $\in [-1, 1]$
- No fine-tuning required

Interpretation:

Score	Meaning
> 0.05	Positive
$[-0.05, 0.05]$	Neutral
< -0.05	Negative

Target-Oriented Sentiment Classification (TSC)

Adapting the framework by Elbouanani et al. (2025), we measure political bias as **prediction inconsistency** across a temporal dimension to track how biases evolve over model iterations.

1. Data Selection & Control

- **Entity Set (E):** $N = 250$ political figures from diverse geopolitical regions, balanced across the political spectrum (Far-Left to Far-Right) and demographics (Gender, Ethnicity, Age).
- **Sentence Set (S):** 60 news-style templates (Pos/Neg/Neu).
- **Counterfactual Consistency:** Explicit removal of role-specific identifiers (e.g., “The President”) ensures the entity name is the only varying factor.
- **Linguistic Adaptation:** Cross-testing in English and Chinese to decouple corpus-driven bias from model architecture.

Implicit Consistency (IC) Score

Entity Replacement and Entropy-Based Bias:

To calculate IC, we first perform entity replacement for each neutral sentence s_i to generate a prediction set:

$$Y_i = \{p(e_j, s_i) \mid e_j \in \mathcal{E}\}$$

We then derive the probability distribution for each sentiment label

$$c \in \{\text{pos}, \text{neg}, \text{neu}\}$$

using:

$$P(c) = \frac{\text{count}(c)}{n}$$

These probabilities are used to compute the Shannon entropy for that specific sentence:

$$H(Y_i) = - \sum_{c \in \{\text{pos}, \text{neg}, \text{neu}\}} P(c) \log P(c)$$

Finally, the global bias score is determined by averaging the entropy values across all m sentences:

$$IC = \frac{1}{m} \sum_{i=1}^m H(Y_i)$$

Temporal Bias Dynamics Metrics

1. Bias Velocity (β):

Is bias increasing or decreasing across versions?

$$\beta = \frac{IC(V_n) - IC(V_{n-1})}{\Delta t}$$

Interpretation:

- $\beta < 0 \rightarrow$ bias decreasing (more neutral)
- $\beta > 0 \rightarrow$ bias increasing (more opinionated)

2. Alignment Delta (Δ_{aln}):

Does alignment make the model more or less biased?

$$\Delta_{aln} = IC(\text{Chat}) - IC(\text{Base})$$

Interpretation:

- $\Delta_{aln} < 0 \rightarrow$ alignment reduces bias
- $\Delta_{aln} > 0 \rightarrow$ alignment increases bias
- Chat model = aligned (with human feedback)
- Base model = raw (no alignment)

Primary Metrics:

- **Bias Score** (B_i): Main measure
- **Variance** (σ^2): Consistency
- **Bias Reduction**: Pre vs. Post

Bias Reduction Formula:

$$\Delta B = \frac{|B_{\text{pre}}| - |B_{\text{post}}|}{|B_{\text{pre}}|} \times 100\% \quad (4)$$

Statistical Tests:

- **T-test**: Pre vs. Post comparison
- **ANOVA**: Cross-model comparison
- **Effect Size**: Cohen's d

Validation:

- Cross-validation with multiple runs
- Manual annotation sample
- Comparison with prior studies

We expect the work to contribute as follows:

- We conceptualize political bias in LLMs as a dynamic property, rather than a static characteristic.
- We provide a longitudinal analysis of political bias across model generations using complementary stance- and sentiment-based metrics.
- We empirically link observed bias shifts to documented alignment and safety interventions, contributing to a deeper understanding of value alignment over time.

Key Contributions:

- 1 Systematic methodology for measuring political bias in LLMs
- 2 Comprehensive comparison across 7 model families from 5+ regions
- 3 Pre vs. Post training analysis
- 4 Open-source pipeline for reproducibility

References

- Buyl, M., et al. (2024). "Large language models reflect the ideology of their creators." *Nature Machine Intelligence*.
- Röttger, P., et al. (2024). "Political compass or spinning arrow? Towards more meaningful evaluations for values and opinions in large language models." *ACL 2024*.
- Feng, S., et al. (2023). "From pretraining data to language models to downstream tasks: Tracking the trails of political biases." *ACL 2023*.
- Hartmann, J., et al. (2023). "The political ideology of conversational AI: Converging evidence on ChatGPT's pro-environmental, left-libertarian orientation."
- Hutto, C.J. & Gilbert, E.E. (2014). "VADER: A Parsimonious Rule-based Model for Sentiment Analysis." *ICWSM*.

Thank You!

Questions?

Repository: <https://huggingface.co/spaces/moujar/TEMPO-BIAS> & GitHub:
<https://github.com/moujar/TEMPO-BIAS>